

1. Introduction to PHP

1. Why is PHP called a loosely typed (dynamically typed) language? Give an example where this causes unexpected behaviour.
2. Explain the difference between == and ===, and between != and !==. What does PHP do behind the scenes in each case?
3. Explain the difference between single quotes and double quotes when defining a string. Which one supports variable interpolation?
4. What is the difference between echo and print?
5. Explain variable scope in PHP: local, global.
6. What is the difference between define() and the const keyword when declaring constants?
7. What is the difference between include, include_once, require, and require_once? What happens in each case when the file is missing?
8. Write one working example for each of the functions below. For each, state what it returns and what data type that return value is.

String functions strlen(), str_word_count(), str_contains(), strpos(), strtoupper(), strtolower(), ucfirst(), ucwords(), str_replace(), strrev(), trim(), explode(), implode(), substr(), sprintf(), nl2br()

Number / type-checking functions is_int(), is_float(), is_nan(), is_numeric(), is_string(), is_array(), is_bool(), gettype(), intval(), round(), floor(), ceil(), abs(), rand()

Date and time functions date(), strtotime(), time(), mktime(), date_default_timezone_set(), date_default_timezone_get()

Array functions array(), array_keys(), array_values(), array_merge(), array_push(), array_pop(), array_shift(), array_unshift(), array_reverse(), array_slice(), array_search(), in_array(), array_key_exists(), array_map(), array_filter(), array_unique(), sizeof(), count(), sort(), rsort(), asort(), ksort(), usort()

Other define(), include, require, json_encode(), json_decode(), die(), exit(), header(), var_dump().

12. What is the difference between count() and sizeof()? Is there any real difference?
13. What is the difference between sort(), asort(), and ksort()? Which ones preserve keys?
14. What is var_dump()?
15. By default, json_decode() returns an object. How do you make it return an associative array instead?
16. Describe the superglobal variables in PHP. List all of them and explain the purpose of each: \$GLOBALS, \$_SERVER, \$_GET, \$_POST, \$_REQUEST, \$_FILES, \$_ENV, \$_COOKIE, \$_SESSION.
17. Why must header() be called before any output is sent to the browser?
18. What is the difference between die()/exit() and simply return?

19. Explain the difference between an indexed array, an associative array, and a multidimensional array, with an example of each.
20. What is a ternary operator, a null coalescing operator (??), and a null coalescing assignment operator (??=)? Give an example of each.

2. PHP Form Validation

1. Differentiate between the HTTP GET and POST methods.
2. When should you use \$_REQUEST? Why is it usually discouraged?
3. Explain the enctype attribute of an HTML form. What are the three possible values, and when must you use multipart/form-data?
4. What do you understand by field sanitization? How is it different from validation?
5. What does htmlspecialchars() actually do to your input value? Which characters does it convert, and which attack does it prevent?
6. Explain the following functions with an example, and clearly state the difference between them: isset(), empty(), is_null(). What does each return for "", "0", 0, null, and an undefined variable?
7. What are filter_var() and the filter constants (FILTER_VALIDATE_EMAIL, FILTER_VALIDATE_INT, FILTER_SANITIZE_*)? Write a validation example using them.
8. What is the difference between client-side validation (HTML5/JavaScript) and server-side validation? Why can you never rely on client-side validation alone?
9. Explain the \$_FILES array.
10. How do you securely handle a file upload? Discuss move_uploaded_file(), is_uploaded_file(), checking the MIME type, restricting the extension, limiting the size, and renaming the file.

3. PHP Session and Cookie

1. Compare \$_SESSION and \$_COOKIE in terms of storage location, size limit, lifetime, and who can read them.
2. What will happen if you do not set a cookie's expiry time? How long has such a cookie survive?
3. How does session_start() work internally?
4. What happens if you try to use \$_SESSION without calling session_start() first?
5. Why must session_start() be called before any HTML output?
6. Explain the full signature of setcookie(). What do the path, domain, secure, and httponly parameters do?
7. How do you delete a cookie?
8. Why is a cookie you just set with setcookie() not available in \$_COOKIE on the same page load?

9. What is the difference between `session_unset()`, `session_destroy()`?
10. How would you implement a "Remember Me" login feature securely using cookies?
11. How can sessions work if the user has cookies disabled?
12. Write a simple login/logout flow using sessions: login page, a protected page that redirects unauthenticated users, and a logout script.

4. PHP MySQL

1. What are the differences between MySQLi and PDO? Explain scenarios where you should use PDO over MySQLi and vice versa.
2. Explain the difference between the procedural and object-oriented styles of MySQLi with an example of each.
3. What is SQL injection? Show a vulnerable query and then show how prepared statements fix it.
4. What is a prepared statement? Explain the difference between `bindParam()`, `bindValue()`, and passing an array to `execute()`.
5. What is the difference between `mysqli_fetch_assoc()`, `mysqli_fetch_array()`, and `mysqli_fetch_object()`?

5. PHP MVC

1. Explain the MVC architecture. What is the responsibility of the Model, the View, and the Controller?
2. Draw or describe the flow of a single request through an MVC application, from the browser to the response.
3. What problems does MVC solve compared to writing all logic in a single PHP file?
4. What is the front controller pattern? Why does almost every MVC framework route all requests through a single `index.php`?

6. PHP AJAX

1. Define a synchronous request versus an asynchronous request. What does the user experience differ between the two?
2. What does AJAX stand for?
3. Describe the `readyState` values and status codes of `XMLHttpRequest`. Which combination signals a successfully completed request?
4. What is the difference between sending data as `application/x-www-form-urlencoded`, `multipart/form-data`, and `application/json`?